

Day: _____

Assignment No 2

Date: _____

Name

Laiba Mudassar

Roll No

SP26-BCS - 021

Course

Applied physics (Theory)

Submitted to:

Mam Ifrah

COMSATS University
Vehari

Question No 1

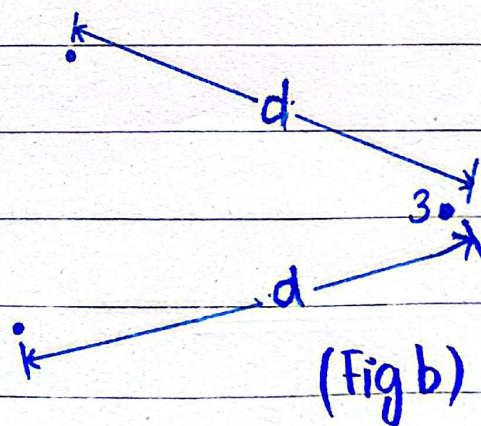
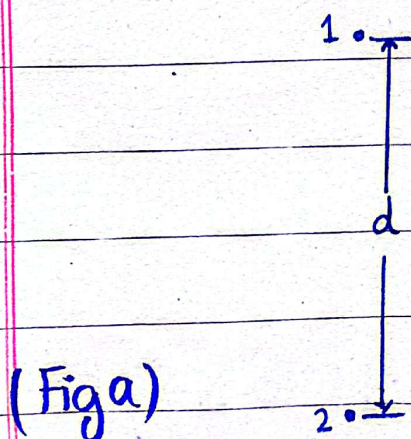
(A)

Calculate the magnitude when Fig(a), two particles (1 and 2) each have a charge of (particle) $+20.0$ and are held at separation distance $d = 1.50\text{m}$.

(a) What is magnitude of electrostatic force on particle 1 due to 2.

In Fig (b), a third particle (3) with charge $+20.0\text{mC}$ is placed so that the three particles form an equilateral triangle of side d .

(b) What is magnitude of electrostatic force on particle 1 due to particle 2 and 3?



Solution: (Part A Q No 1)(a)Given data:

We are given two particles both with charge $q_1 = +20.0 \mu\text{C} = +20.0 \times 10^{-6} \text{C}$

$$q_2 = +20.0 \mu\text{C} = +20.0 \times 10^{-6} \text{C}$$

$$\text{distance}(d) = r = 1.50 \text{m}$$

To Find :

Magnitude of electrostatic force on particle 1 due to particle 2 ?

Formula:

To find magnitude of electrostatic force, Use Coulomb's law,

$$F = k \frac{|q_1 q_2|}{r^2}$$

Put value in this formula,

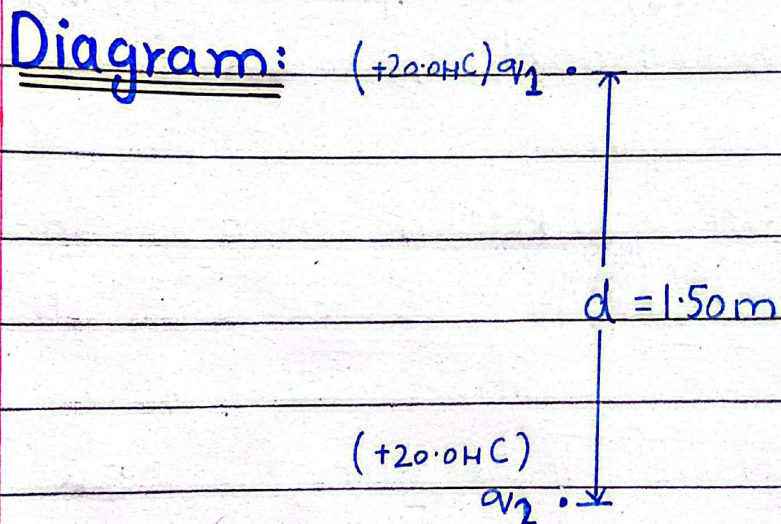
We know:

$$\text{Coulomb's constant } k = 8.99 \times 10^9 \text{ Nm}^2/\text{C}^2$$

$$F_{12} = \frac{(8.99 \times 10^9)(20.0 \times 10^{-6})(20.0 \times 10^{-6})}{(1.50)^2}$$

$$F_{12} = 1.6 \text{ N}$$

So magnitude of electrostatic force on particle 1 due to particle 2 is 1.6 N



(b)

Net force in equilateral triangle

A third particle with same charge $+20.0 \mu\text{C}$ is added from an equilateral triangle with side d .

To Find:

Net force F_{net} on particle 1.

Solution:

Individual Magnitudes:

Since all charges and distance are equal, the force from particle 2 (F_{12}) and particle 3 (F_{13}) are both 1.60N . F_{13} mean force on 1 due to 3 particle.

Direction: In equilateral triangle, the internal angle is 60° . Because all charges are positive, the force are repulsive. The angle between the two force vector acting on particle 1 is 60° .

Net force (F_{net}):

Using Vector addition,

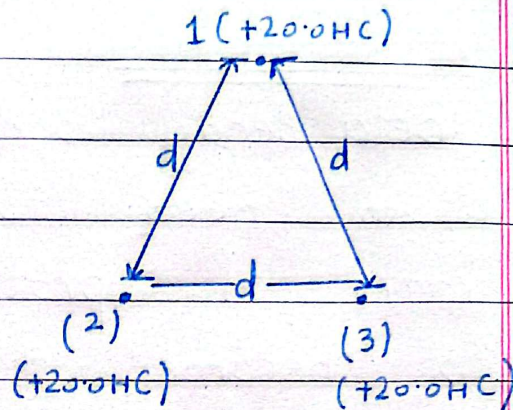
$$F_{\text{net}} = \sqrt{F_{12}^2 + F_{13}^2 + 2F_{12}F_{13}\cos(60^\circ)}$$

Put values in Formula,

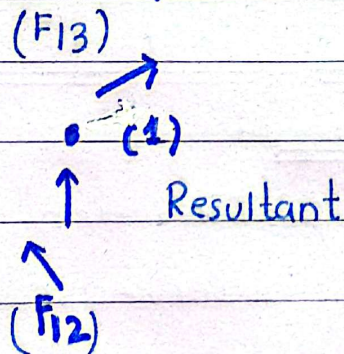
$$F_{\text{net}} = \sqrt{(1.6)^2 + (1.6)^2 + 2(1.6)(1.6)(0.5)}$$

$$F_{\text{net}} = \sqrt{7.68}$$

$$F_{\text{net}} = 2.77 \text{ N}$$



Forces on particle 1



Question 1(Part B)

Calculate the type of particle (electron or positron) produced in each beta decay process:

(a) When a proton transform into a neutron,

(b) When a neutron transform into proton.

Answer:

Beta-Decay:

Beta decay occurs when an atom need to change its ratio of protons to neutrons to become more stable.

(a)
Proton transforms into a
neutron, (β^+ decay):

This occur in "proton-rich" nuclei that need to decrease their proton count to reach stability.

Process:

A proton inside the nucleus transform into to neutron.

Charge balance:

Since proton is positive (+1)

and a neutron is neutral (0), the positive charge carried must away from nucleus.

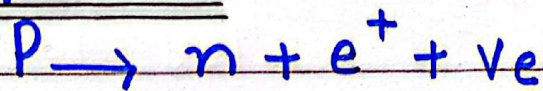
Particle produced:

A positron (also called β^+ particle) is ejected.

A positron is antimatter version of electron - it has same mass but positive charge.

A neutrino (ν_e) is also emitted.

Equation:



(e^+) $\rightarrow$ is positron

Result:

The atomic number (Z) decreases by 1, changing into a different element, while the mass (A) remain same.

(b)

Neutron transforms into
proton:
(β^- -Decay)

This occurs in "neutron rich" nuclei that need to decrease their neutron count to reach stability.

Process:

A neutron inside the nucleus transforms into proton.

Charge balance:

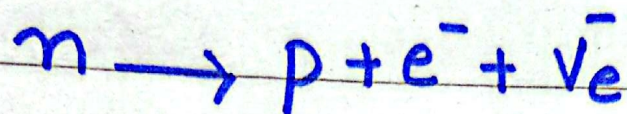
Since a neutron is neutral (0), and proton is positive (+1), the laws of physics require a negative charge to be created to keep net charge at zero.

Particles produced:

An electron (also called β^-) particle is ejected from nucleus at high speed.

An additional particle antineutrino ($\bar{\nu}_e$) is also emitted for conservation of energy and momentum.

Equation:



$e^- \rightarrow$ produced
electron.

$\bar{\nu}_e \rightarrow$ antineutrino

Result: Atomic number (Z)

increase by 1, changing it
into a different element,
but mass(A) stays same.
